What Bars Cannot See: Order Flow, Its OHLCV Proxies, and Short-Horizon Bitcoin Returns

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Abstract

Microstructure variables built from open-high-low-close-volume (OHLCV) bars, such as bulk volume classification (BVC), Kyle's lambda and high-low spread estimators, are widely used when trade-level data are unavailable. We test what they measure and what they predict, using almost five years of hourly and one-minute Binance bars for BTC/USDT and ETH/USDT in which the exchange reports the aggressor-side (taker) volume of every bar, so the true signed order flow is observed. First, range-based volatility is measured almost perfectly (correlation 0.99 with one-minute realised volatility), but signed flow is not: hourly BVC imbalance has a correlation of 0.49 with observed imbalance, and a nonparametric bound shows that no function of the bar's own return explains more than about a quarter of its variance (0.25). Spread estimators are dominated by volatility, with medians four orders of magnitude above the one-tick minimum spread. These results are nearly identical for both assets. Second, in BTC, a one standard deviation taker-imbalance shock is followed by a price decline of about 1.3 basis points over two hours ($t = -4.0$), robust to return, volume and volatility controls and stable across sub-samples. This information lies in the part of flow that leaves no trace in the hourly bar: predictive content rises with measurement accuracy, from none for hourly BVC, to part for BVC on one-minute bars, to all for observed flow, which subsumes the proxy. Used alone in a monthly walk-forward, observed imbalance forecasts BTC returns out of sample (Clark-West $p = 0.0001$); in ETH the effect is weaker and not significant. Third, the information is economically negligible. Its break-even trading cost is about one basis point; no multivariate model on microstructure proxies, observed flow or realised measures beats a zero forecast over 53 leakage-controlled test months; and the best of 16 trading rules is insignificant once selection is accounted for (deflated Sharpe ratio 0.70). We also document that a zero-fee promotion multiplied BTC trade counts fivefold while collapsing the dispersion of aggressor imbalance, a warning for research using fee-free periods.

Keywords: order flow, bulk volume classification, liquidity proxies, cryptocurrency, return predictability, out-of-sample evaluation, transaction costs
JEL: G12, G14, G17, C53, C58

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Code and data access: https://github.com/a0x0h/what-bars-cannot-see.

1 Introduction

Microstructure theory ties short-horizon price changes to the arrival of signed order flow. Informed traders reveal information through the direction of their trades (Kyle, 1985; Glosten and Milgrom, 1985), liquidity providers set prices to manage inventory and adverse selection, and order flow has a price impact that is partly permanent and partly transient (Bouchaud et al., 2018). Testing these ideas requires knowing which side initiated each trade. Because such data are expensive or unavailable for many markets and periods, a large applied literature reconstructs microstructure quantities from open-high-low-close-volume (OHLCV) bars. Spreads are estimated from high and low prices (Corwin and Schultz, 2012; Abdi and Ranaldo, 2017), illiquidity from returns and volume (Amihud, 2002), and signed flow by bulk volume classification (BVC), which splits each bar's volume between buyers and sellers according to the standardised price change (Easley et al., 2012, 2016). These variables are increasingly used as features in machine-learning forecasts of returns, including in cryptocurrency markets (Easley et al., 2021, 2024).

Two questions are rarely asked together. How much of the underlying quantity do bar-based proxies capture? And does what they capture, or the true quantity itself, predict returns once the evaluation is free of look-ahead and the signals are priced at realistic costs? We answer both in a setting where the ground truth is observed. Binance, the largest cryptocurrency spot exchange, publishes for every bar the volume executed by buyer-initiated (taker) orders. The signed order flow of every hour and every minute is therefore known. We verify on one month of trade-level data that the bar-level taker volume reproduces trade-level aggressor flow exactly (correlation 1.0000 over 720 hours). We use hourly and one-minute bars for BTC/USDT and ETH/USDT from January 2022 to October 2026, about 42,000 hours and 2.5 million minutes per asset.

Measurement. We score each proxy against its observed counterpart. Range-based volatility from hourly bars has a correlation of 0.99 with realised volatility from one-minute returns, so bar data measure volatility well. Signed flow is another matter. Hourly BVC imbalance has a correlation of 0.49 with observed taker imbalance and the correct sign in 72% of hours; summing BVC over one-minute bars raises the correlation to 0.60. Neither the choice of volatility window nor a heavy-tailed distribution helps. The reason is not the functional form. BVC and the tick rule are functions of the bar's standardised return, and we show nonparametrically that *no* such function explains more than 0.25 of the variance of hourly taker imbalance (0.34 on one-minute bars), even in sample. BVC is already close to this bound, so better calibration cannot help much, and its scale drifts over time. Most aggressor flow is absorbed without moving the bar's price. Kyle's lambda computed from BVC-signed volume inherits the error and has a correlation of only 0.26 with lambda from observed flow. Spread estimators fare worst: on a market whose minimum (one-tick) spread is about 0.002 basis points, the Corwin-Schultz estimate has a median of 15.8 basis points and moves with volatility (correlation 0.87), the extreme case of the bias described by Jahan-Parvar and Zikes (2023).

Information. In BTC, observed flow is informative about future prices, but in a specific way. A one standard deviation increase in hourly taker imbalance is followed by a price decline of 1.3 basis points over the next two hours ($t = -4.0$) that fades within a day, the signature of transient

price impact or of informed passive liquidity. The reversal is of similar size under standard fees and in both halves of the sample, and it survives controls for lagged returns, momentum, volume and volatility. Holding the bar's own return fixed, the coefficient on observed imbalance is unchanged, while the return itself has no predictive power. The information in order flow therefore resides in the part that leaves no trace in the hourly bar's price, which hourly BVC cannot recover and which, consistently, it does not predict. BVC on one-minute bars recovers part of the signal, and observed flow subsumes it in a direct comparison: predictive content rises with measurement accuracy. Where flow is informative, measurement error in the proxy is not innocuous. In ETH the reversal has the same sign in most sub-samples but is about half as large and not significant.

Prediction and economic value. The effect is statistically detectable but economically tiny, explaining about 0.016% of the variance of four-hour returns. We confirm this in a monthly walk-forward evaluation in which every model is re-estimated on purged training data, preprocessing and hyper-parameters are chosen inside each training window, and automated tests verify that no feature uses future data. Over 53 out-of-sample months, none of 8 elastic-net and gradient-boosting specifications for BTC, built on technical indicators, microstructure proxies, observed flow, realised measures or all of them, beats the naive zero forecast in mean squared error. Clark-West tests detect that microstructure variables add some information to technical indicators in nested comparisons, but the gains are too small to survive estimation error, and cross-validation inside each training window selects the empty model in most months. For ETH, and for BTC at horizons of one, eight and 24 hours, a technical-indicator model achieves a small positive out-of-sample R^2 of about a tenth of a percent, and adding microstructure variables never improves on it. A one-variable model with observed taker imbalance alone, by contrast, has a small but significant out-of-sample R^2 for BTC (0.021% at two hours, Clark-West $p = 0.0001$), while the same model with hourly BVC imbalance has none: the signal is detectable out of sample when flow is measured and absent when it is proxied from hourly bars, and multivariate shrinkage on a short validation block discards it. Traded on its own, it breaks even at about one basis point per unit of turnover. Turning the multivariate forecasts into an overlapping-sleeve trading rule, the best of 16 model and rule combinations for BTC has a gross Sharpe ratio of 1.24 and a break-even cost of 9.0 basis points, but its deflated Sharpe ratio (Bailey and López de Prado, 2014) is 0.70 and its sign changes across years.

Contributions. First, we provide what is, to our knowledge, the longest direct validation of bar-based order-flow, price-impact, volatility and spread proxies against exchange-reported aggressor flow, and an information ceiling that applies to every return-based classifier, not only to BVC. Second, we show that, where order flow predicts intraday returns, its predictive content lies in the component that bar-based proxies cannot see, which links the measurement and the prediction literatures. Third, we document a natural experiment in data quality: during Binance's zero-fee promotion on BTC/USDT, hourly trade counts rose more than fivefold while the dispersion of aggressor imbalance fell by four-fifths, consistent with costless round-trip trading, and ETH/USDT, which was not included, was unaffected. Fourth, we release a fully reproducible pipeline with leakage tests, so the negative forecasting results can be checked rather than taken on trust.

The rest of the paper is organised as follows. Section 2 reviews related work. Section 3 describes the data and the institutional setting, Section 4 defines the measures, and Section 5 validates the

proxies. Section 6 studies what observed and proxy flow tell us about future prices. Section 7 reports the out-of-sample forecasting and trading results, Section 8 the robustness checks, and Section 10 concludes.

2 Related literature

Do low-frequency liquidity measures measure liquidity? A long line of work benchmarks spread and price-impact proxies built from daily data against estimates from intraday trades and quotes. Goyenko et al. (2009) establish the approach for U.S. equities; Corwin and Schultz (2012) and Abdi and Ranaldo (2017) propose the high-low and close-high-low estimators that now dominate applied work. Jahan-Parvar and Zikes (2023) show that several of these estimators carry a volatility-induced bias that dominates when the true spread is small relative to volatility, and that the bias has grown as spreads have narrowed. In cryptocurrency markets, Brauneis et al. (2021) compare low-frequency measures with order-book benchmarks for bitcoin and ether on three exchanges and find that the Corwin-Schultz and Abdi-Ranaldo estimators track time variation in liquidity best, while Amihud and Kyle-Obizhaeva measures capture levels. We add to this literature in two ways. First, we score proxies of *signed order flow*, not only of spreads, against an exchange-reported ground truth over almost five years. Second, we ask whether the measurement error matters for the downstream use that motivates many applications, namely short-horizon return forecasting.

Trade classification and flow toxicity. When aggressor flags are unavailable, trades are signed by the tick rule or by Lee and Ready (1991). Easley et al. (2012, 2016) propose bulk volume classification (BVC), which assigns a fraction $\Phi(\Delta p/\sigma)$ of each bar's volume to buyers, and argue that it captures the information content of order flow better than trade-level rules. Evidence on its accuracy is mixed. Chakrabarty et al. (2015) find that BVC is less accurate than the tick rule and Lee-Ready for individual equities, Panayides et al. (2019) find that it identifies aggressors less accurately but is more closely related to informed trading and improves after calibration, and Andersen and Bondarenko (2014) question whether VPIN built on BVC predicts toxicity events. These studies use equity or futures data over comparatively short samples. We exploit the fact that Binance reports the taker-buy volume of every bar to evaluate BVC on hourly and one-minute bars over about 42,000 hours per asset, and to bound how much of the true flow any return-based classifier can recover.

Order flow and short-horizon returns. Order imbalance predicts short-horizon index returns (Chordia et al., 2002), order-flow imbalance at the top of the book explains contemporaneous price changes (Cont et al., 2014), and deep-learning models extract predictive signals from order-book flow at horizons of seconds to minutes (Kolm et al., 2023). For cryptocurrencies, Makarov and Schoar (2020) document that signed order flow explains a large share of bitcoin returns across exchanges, Aleti and Mizrach (2021) compare spot and futures microstructure, and Easley et al. (2024), following Easley et al. (2021), report that microstructure measures such as the Roll measure and VPIN predict future crypto market dynamics in a machine-learning framework. Time-series predictability at horizons of days to weeks is documented by Liu and Tsyvinski (2021). Our study

differs from this work in focusing on the intraday horizon at which microstructure effects should be strongest, in measuring predictive gains strictly out of sample with tests designed for nested and overlapping forecasts (Clark and West, 2007; Campbell and Thompson, 2008), and in pricing the resulting signals at realistic transaction costs. Because a single dataset invites specification search, we fix the design before estimation, report every feature set and model we estimate, and interpret Sharpe ratios with the caution recommended by Bailey and López de Prado (2014) and Harvey et al. (2016).

3 Data and institutional setting

Source. We use Binance spot klines for BTC/USDT and ETH/USDT at one-hour and one-minute frequency, downloaded from the exchange's public data archive (`data.binance.vision`). Each bar reports open, high, low and close prices, base-asset volume V_t, the number of trades, and the base volume of trades whose taker was the buyer, B_t. The observed signed volume of a bar is $2B_t - V_t$ and the observed imbalance is $(2B_t - V_t)/V_t \in [-1, 1]$. Timestamps are in UTC. The archive switched from millisecond to microsecond resolution in January 2025, and we normalise all timestamps. We remove bars with non-positive prices or a close outside the high-low range and do not fill missing bars; rolling statistics that would span a gap are set to missing. The sample runs from 1 January 2022 to 31 October 2026 and contains one missing hour per asset.

Ground truth. The validity of the exercise rests on the taker volume being the true aggressor-side volume. We verify this for September 2026 by downloading every aggregated trade of BTC/USDT, signing each by its buyer-maker flag, and summing to the hour. The hourly imbalance computed from trades and from klines has a correlation of 1.0000 and a mean absolute difference of 0.00011 over 720 hours.

Tick size and fees. Both pairs trade with a tick of 0.01 USDT, about 0.002 basis points for BTC and 0.043 for ETH at median sample prices, which is the minimum possible quoted spread. The standard spot fee for regular accounts is 10 basis points per side, reduced for high-volume tiers and fee discounts. From 8 July 2022 to 22 March 2023 Binance waived all fees on BTC/USDT and other BTC pairs, but not on ETH/USDT. Figure 1 shows the effect. The median number of trades per hour on BTC/USDT rose from 40,984 in the first half of 2022 to 221,794 during the promotion and fell to 28,844 after it ended, with median hourly dollar volume moving from \$60 million to \$178 million and back to \$32 million. On ETH/USDT the median trade count was 28,226, 27,976 and 14,727 over the same windows. Zero fees make round-trip and self-matching trades costless, which is the setting in which Cong et al. (2023) document wash trading on unregulated venues. We therefore treat the promotion as a separate regime throughout.

Descriptive statistics. Table 1 summarises the two series. Hourly log returns have a standard deviation of about 54 basis points for BTC and 71 for ETH, with fat tails. Observed taker imbalance has a mean close to zero, a standard deviation of 0.13 and modest persistence. Its correlation with the same hour's return is only 0.39 for both assets, an early sign that prices reveal only part of the flow.

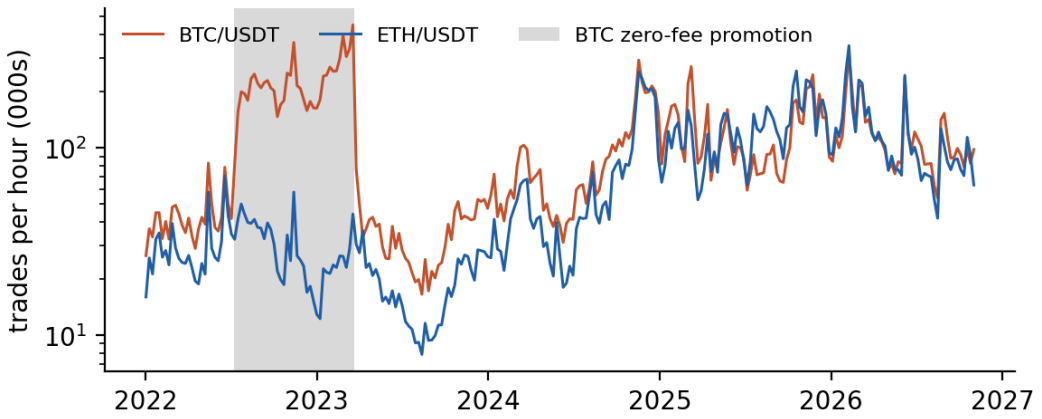


Figure 1: Hourly trade counts, weekly medians (log scale). The shaded window is Binance's zero-fee promotion on BTC pairs (8 July 2022 to 22 March 2023); ETH/USDT was not included.

Table 1: Descriptive statistics, hourly bars

Asset	Start	End	Hours	Missing	$\sigma(r_{1h})$	$\sigma(r_{4h})$	Kurt(r_{1h})	Med. \$vol (M)	Med. trades	Mean imb.	SD imb.	AC(1) imb.	Corr(imb., r_{1h})
BTC/USDT	2022-01-01	2026-10-31	42,359	1	53.5	105.8	13.7	60.3	85,699	−0.021	0.130	0.18	0.39
ETH/USDT	2022-01-01	2026-10-31	42,359	1	71.2	142.3	16.3	31.1	48,442	−0.009	0.126	0.21	0.39

$\sigma(r)$: standard deviation of hourly and four-hour log returns in basis points. \$vol: quote (USDT) volume per hour. Imb.: observed taker imbalance $(2B_t - V_t)/V_t$; AC(1) is its first-order autocorrelation.

4 Measures

All measures at bar t are functions of bars up to and including t; Appendix Table 12 lists definitions and windows. We group them by the quantity they target.

Spread and price impact (proxy). The two-bar Corwin and Schultz (2012) high-low estimator averaged over 24 hours; a causal version of the Abdi and Ranaldo (2017) close-high-low estimator that uses bars $t-1$ and t only; the Roll (1984) estimator over 48 hours; the Amihud (2002) ratio; and a square-root impact measure $|r_t|/\sqrt{V_t/\bar{V}}$ (Bouchaud et al., 2018).

Order flow (proxy). Following Easley et al. (2016), the buy fraction of bar t is $\Phi(r_t/\hat{\sigma}_t)$, where $\hat{\sigma}_t$ is the trailing 168-hour standard deviation of hourly returns, so the BVC signed volume is $(2\Phi(r_t/\hat{\sigma}_t) - 1)V_t$. From it we build imbalance at one and 24 hours, its acceleration (6-hour minus 24-hour imbalance), a VPIN-style toxicity measure (the 24-hour sum of absolute signed volume over total volume), and Kyle's lambda from a rolling 48-hour regression of returns on signed dollar volume.

Observed flow and trade activity. The same five constructions applied to observed signed volume $2B_t - V_t$, which gives matched pairs of proxy and observed features; standardised trade counts and average trade size; and the taker imbalance of the last 15 minutes of the hour.

Volatility and intrahour measures. Range-based estimators of Garman and Klass (1980), Rogers and Satchell (1991) and Yang and Zhang (2000) over 24 hours, the ratio of 24-hour to 168-hour

volatility and volatility of volatility. From one-minute bars of the same hour we compute realised variance, the jump share $\max(RV - BV, 0)/RV$ with bipower variation BV (Barndorff-Nielsen and Shephard, 2004), and realised skewness (Amaya et al., 2015).

Efficiency, bar structure and activity. Efficiency ratio, return autocorrelation, variance ratio, candle body and wick shares, and standardised volume.

Technical controls. Momentum over 1, 4, 24 and 168 hours, RSI(14), MACD histogram, Bollinger %B, gaps to the 24- and 168-hour moving averages, and ATR(14).

Feature sets. TI contains the 10 technical controls; MS_{proxy} the 23 OHLCV-only microstructure measures; MS adds observed flow, trade activity and one-minute measures (35 features); ALL is MS plus TI (45 features). Flow_{BVC} and $\text{Flow}_{\text{taker}}$ are the five matched flow constructions from BVC and from observed volume. The first four sets are nested, which the incremental tests exploit; the two flow sets differ only in the input.

5 Do bar-based proxies measure what they claim?

This section scores the OHLCV-only measures against the quantities they are meant to capture. Observed signed flow is the taker imbalance $(2B_t - V_t)/V_t$; the benchmark for volatility is realised variance from one-minute returns. Because quoted spreads are not in the public archive, we evaluate the spread estimators indirectly, against the tick size and against volatility. Table 2 collects the main statistics for BTC/USDT; ETH/USDT results are in Appendix Table 23.

Signed order flow. Hourly BVC imbalance has a correlation of 0.49 with observed taker imbalance and the same sign in 72% of hours. Aggregating to 24 hours barely changes the correlation (0.51), so the error is not high-frequency noise that averages out. Applying BVC to one-minute bars and summing within the hour raises the correlation to 0.60, the largest gain available from the data we use. The choice of the volatility window (24 hours to 30 days) or of a heavy-tailed Student-t CDF moves the correlation by at most two hundredths (Appendix Table 13). The tick rule, which signs the whole bar by its return, is worse on hourly bars but nearly as good as BVC on one-minute bars, consistent with Chakrabarty et al. (2015) and Panayides et al. (2019).

An information ceiling for return-based classification. BVC, the tick rule and their variants are all functions of the standardised bar return $z_t = r_t/\hat{\sigma}_t$. Their accuracy is therefore bounded by how much of the observed imbalance is explained by $\mathbb{E}[\text{imb}_t \mid z_t]$, whatever the functional form. We estimate this conditional mean nonparametrically with 50 quantile bins of z_t. As an upper bound we compute its in-sample R^2; with 50 bins and thousands of hours per period the overfitting bias is negligible. Over the full sample the bound is 0.25 on hourly bars and 0.34 on one-minute bars, and in no calendar year does the hourly bound exceed 0.33 (Appendix Table 14). The squared correlation of BVC with observed imbalance is 0.24, almost exactly at the bound: the shape of the BVC map is close to optimal, and no other function of the bar return would do materially better.

Table 2: Bar-based proxies against observed quantities, BTC/USDT

	Pearson	Spearman	Same sign	N
Panel A. Signed order flow: proxy vs. observed taker imbalance				
BVC, 1h bars, per hour	0.489	0.541	0.716	42,316
BVC, 1h bars, 4h sums	0.475	0.524	0.701	42,310
BVC, 1h bars, 24h sums	0.507	0.538	0.693	42,270
BVC, 1m bars summed to 1h	0.603	0.637	0.747	42,359
Tick rule, 1h bars, per hour	0.430	0.389	0.716	42,357
Panel B. Price impact and volatility				
Kyle λ (48h): BVC vs. observed flow	0.260	0.213	–	41,978
log GK vol (1h bars) vs. log RV (1m), 24h	0.987	0.983	–	41,978
Panel C. Information ceiling of return-based classifiers (out-of-sample R^2)				
	BVC raw	BVC rescaled	Any $f(z)$	N
1h bars	−6.783	0.209	0.213	21,158
1m bars	0.116	0.308	0.328	1,270,544
Panel D. Spread estimators (one tick = 0.002 bps)				
	Median (bps)	Mean (bps)	Share = 0	Corr. log RV
Corwin-Schultz (24h mean)	15.8	17.5	0.0%	0.87
Abdi-Ranaldo (24h)	0.0	9.6	50.2%	0.64
Roll (48h)	15.1	19.7	38.4%	0.52

Panel A: correlation and sign agreement between proxy and observed taker imbalance $(2B_t - V_t)/V_t$, per hour and over rolling 4- and 24-hour sums. Panel B: Kyle's lambda from 48-hour rolling regressions on BVC-signed and on observed signed dollar volume; Garman-Klass volatility from hourly bars against realised volatility from one-minute returns. Panel C: out-of-sample R^2 for observed imbalance of raw BVC, of BVC after a linear rescaling, and of the best function of the standardised bar return (50 quantile bins), all fitted on the first half and evaluated on the second. Panel D: spread estimators in basis points, share of hours truncated at zero, and correlation of the log estimate with log realised volatility.

Fitted on the first half of the sample and evaluated on the second, the out-of-sample ceiling is 0.21, and rescaled BVC attains 0.21. Fitted on the second half and evaluated on the first, however, it falls to 0.09 for BTC (0.23 for ETH), because the scale of the relation drifts: the slope of observed imbalance on BVC rises from 0.065 in 2022 to 0.169 in 2026. A classifier calibrated in one period is therefore miscalibrated in another, so out-of-sample accuracy is lower than the bound.

Figure 2 shows the mechanism. The BVC map $2\Phi(z) - 1$ has the right shape, but observed imbalance moves far less than the map implies (the slope fitted on the first half of the sample is 0.09 on hourly bars), and its dispersion at any given return is wide. Two conclusions follow. First, the shape of the classifier is not the problem; recalibration, which Panayides et al. (2019) recommend, fixes the scale but must be repeated as the scale drifts. Second, about three-quarters of the hourly variation in aggressor imbalance leaves no trace in the bar's own price change. This flow is absorbed by passive liquidity, offset within the hour, or reflected in prices only later, and no classifier based on the hourly bar can recover it. Part of it is visible in the one-minute price path, which is why BVC on one-minute bars does better.

Price impact. Kyle's lambda estimated from BVC-signed dollar volume has a correlation of only 0.26 with the same estimator applied to observed signed volume. Errors in the regressor are

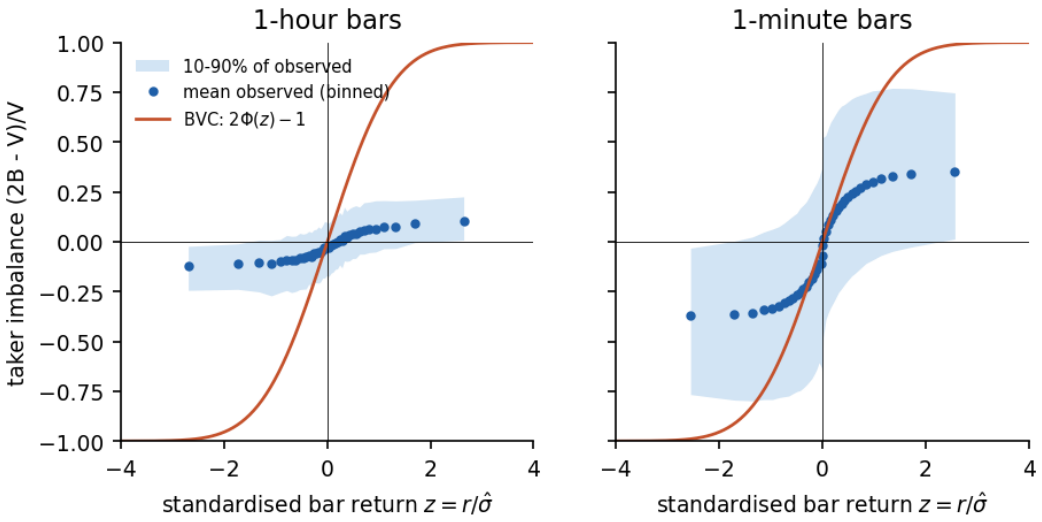


Figure 2: Observed taker imbalance against the standardised bar return. Dots are bin means of observed imbalance in 40 quantile bins of z; the band covers the 10th to 90th percentile within each bin; the line is the BVC mapping $2\Phi(z) - 1$. BTC/USDT, January 2022 to October 2026.

amplified in a ratio of a covariance to a variance, so a proxy that captures about a quarter of the flow variance yields an impact estimate that is largely noise.

Volatility. The range-based Garman-Klass estimator from hourly bars has a correlation of 0.99 with realised variance from one-minute returns (both in logs, 24-hour windows). Bar-based volatility measures are close substitutes for realised measures at this horizon.

Spreads. The minimum quoted spread on BTC/USDT is one tick of 0.01 USDT, about 0.002 basis points at the sample's median price. The 24-hour Corwin-Schultz estimate has a median of 15.8 basis points, so taken at face value it would imply an average book about four orders of magnitude wider than one tick, an implausible value for the most liquid crypto pair. Its logarithm has a correlation of 0.87 with log realised volatility. The Abdi-Ranaldo estimate is truncated at zero in 50% of hours and the Roll estimate in 38%. In this market the spread estimators measure volatility, the extreme case of the bias documented by Jahan-Parvar and Zikes (2023). Liquidity conclusions drawn from them at hourly frequency should be read as volatility conclusions.

Stability. The BVC correlation varies between 0.48 and 0.56 across calendar years and is essentially the same in low-, medium- and high-volatility regimes at the hourly level (Appendix Table 15). It is higher during Binance's zero-fee promotion on BTC/USDT (8 July 2022 to 22 March 2023), when hourly trade counts rose sharply (0.60 against 0.51 under standard fees). The reason is visible in the flow itself. During the promotion the standard deviation of hourly taker imbalance on BTC/USDT fell to 0.030, from 0.140 under standard fees: volume rose fivefold but became almost perfectly two-sided, which is what costless round-trip trading produces. On ETH/USDT, which kept its fees, the dispersion barely changed (0.095 against 0.131). With little genuine directional flow left, the price change of a bar explains a larger share of what remains. The relation between prices and

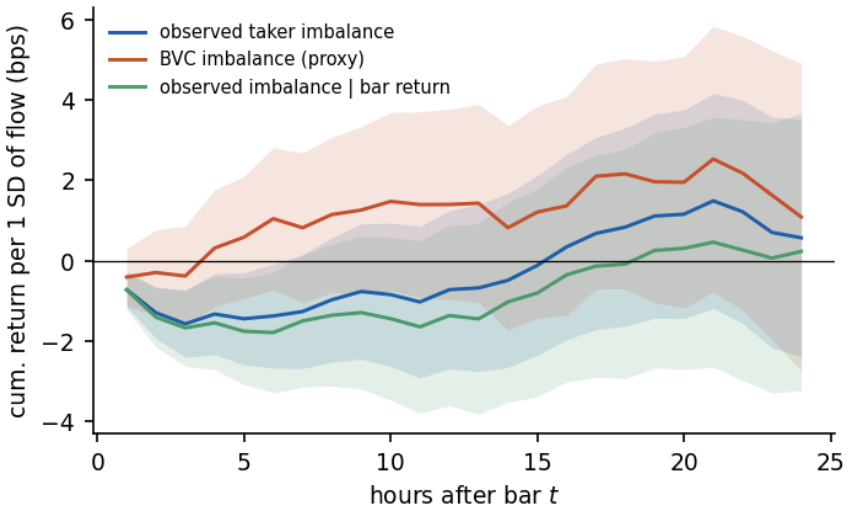


Figure 3: Cumulative log return (basis points) over k hours following a one standard deviation flow shock in hour t, from regression (1), with 95% Newey-West bands. Blue: observed taker imbalance alone. Red: BVC imbalance alone. Green: observed imbalance holding the bar's standardised return fixed. BTC/USDT.

aggressor flow therefore depends on the exchange's fee schedule and on the composition of trading it attracts (Cong et al., 2023), and data from fee-free periods should be treated with care.

6 What does order flow tell us about future prices?

Before turning to out-of-sample forecasting, we ask a simpler question: is the observed flow, or its proxy, related to subsequent returns at all, and if so, through which component? For each horizon $k = 1, \ldots, 24$ hours we estimate

$$\log \frac{C_{t+k}}{C_t} \times 10^4 = a_k + b_k \, \tilde{x}_t + c_k \, \tilde{z}_t + e_{t,k}, \tag{1}$$

where $\tilde{x}_t$ is a standardised flow measure of bar t (observed taker imbalance or BVC imbalance) and $\tilde{z}_t$ the standardised bar return $r_t/\hat{\sigma}_t$, which is the only input of BVC. The coefficient b_k is the cumulative return, in basis points, that follows a one standard deviation flow shock. When $\tilde{z}_t$ is included, b_k measures the effect of the component of flow that is orthogonal to the bar's own price change, i.e. the part that no return-based classifier can recover. Standard errors are Newey-West with $\max(24, 2k)$ lags. These regressions use the full sample and are descriptive; Section 7 asks whether anything survives out of sample.

Figure 3 and Table 3 show three facts. First, observed aggressor flow predicts a *reversal*. A one standard deviation increase in taker imbalance is followed by a price decline of 0.7 basis points over the next hour ($t = -3.4$) and 1.3 basis points over two hours ($t = -4.0$); the decline is 1.3 basis points at four hours and has disappeared after about 15 hours. Buying pressure that is absorbed by passive orders is followed by falling prices, which is consistent with a transient component of price impact (Bouchaud et al., 2018) and with passive liquidity providers being, on average, on the right side of the trade.

Table 3: Return response to order-flow shocks, BTC/USDT

Regressor (1 SD)	$k = 1$h	$k = 2$h	$k = 4$h	$k = 8$h	$k = 12$h	$k = 24$h
Observed taker imbalance	−0.72	−1.30	−1.33	−0.97	−0.72	0.57
	(−3.42)	(−3.99)	(−2.58)	(−1.23)	(−0.72)	(0.38)
BVC imbalance	−0.41	−0.30	0.31	1.15	1.40	1.09
	(−1.12)	(−0.55)	(0.42)	(1.17)	(1.16)	(0.56)
Observed imbalance, given z_t	−0.73	−1.41	−1.55	−1.36	−1.37	0.23
	(−2.97)	(−3.74)	(−2.61)	(−1.51)	(−1.20)	(0.13)
Bar return z_t, given imbalance	0.02	0.26	0.50	0.89	1.48	0.78
	(0.06)	(0.46)	(0.64)	(0.84)	(1.15)	(0.38)

Coefficients from regression (1): cumulative log return in basis points from the close of hour t to $t + k$ per one standard deviation of the regressor; Newey-West t-statistics in parentheses. The third and fourth rows come from one regression with both regressors.

Second, the information is in the part of flow that the bar's price does not reveal. Holding the standardised bar return fixed leaves the coefficient on observed imbalance essentially unchanged (−1.4 basis points at two hours, $t = -3.7$), while the bar return itself has no predictive power ($t = 0.5$ at two hours and $t = 0.6$ at four). Third, and as a consequence, hourly BVC imbalance predicts nothing: its coefficient is −0.3 basis points at two hours ($t = -0.5$) and is positive, if insignificant, at longer horizons. Section 5 showed that about three-quarters of the variance of observed imbalance is invisible in the hourly bar's price change; this section shows that the invisible part is the part that matters.

Controls and the best available proxy. Table 4 asks whether the reversal is a disguised return or volume effect. Adding three lags of the hourly return, 24-hour momentum, standardised volume, volatility and the previous hour's imbalance leaves the two-hour coefficient at −1.3 basis points ($t = -3.7$). The flow effect is not a reversal of past returns and not a volume or volatility effect. The last rows use the most accurate proxy we can build from price data, BVC applied to one-minute bars and summed to the hour, whose correlation with observed imbalance is 0.60 rather than 0.49. On its own it does predict a reversal (−1.0 basis points at two hours, $t = -2.5$). When it competes with observed imbalance, however, its coefficient falls to −0.3 ($t = -0.7$) while observed imbalance keeps −1.1 ($t = -2.6$). Predictive content therefore rises with measurement accuracy: none for hourly BVC, part of the signal for one-minute BVC, and all of it for observed flow, which subsumes the proxy.

The magnitude is small. At four hours the observed-imbalance regression explains 0.016% of the return variance, and the implied move of one to two basis points per standard deviation compares with a standard spot fee of ten basis points per side. Even a perfect measurement of flow at this frequency would therefore be of little direct use for trading, a conclusion the next section confirms out of sample.

Table 4: Return response to observed flow with controls and against the best proxy

Specification (coefficient of interest)	$k = 1$h	$k = 2$h	$k = 4$h
BTC/USDT			
(1) taker	−0.72 (−3.4)	−1.30 (−4.0)	−1.33 (−2.6)
(2) + bar return	−0.73 (−3.0)	−1.41 (−3.7)	−1.55 (−2.6)
(3) + lagged returns, 24h momentum	−0.73 (−2.9)	−1.39 (−3.6)	−1.52 (−2.5)
(4) + volume, volatility, lagged taker	−0.64 (−2.5)	−1.32 (−3.7)	−1.52 (−2.8)
(5) 1m BVC alone	−0.65 (−2.4)	−1.00 (−2.5)	−0.75 (−1.3)
(6) taker vs 1m BVC: taker	−0.52 (−1.9)	−1.09 (−2.6)	−1.38 (−2.1)
(6) taker vs 1m BVC: 1m BVC	−0.33 (−1.0)	−0.35 (−0.7)	0.09 (0.1)
ETH/USDT			
(1) taker	0.10 (0.3)	−0.53 (−1.1)	0.55 (0.8)
(2) + bar return	−0.15 (−0.4)	−0.79 (−1.5)	0.21 (0.3)
(3) + lagged returns, 24h momentum	−0.13 (−0.4)	−0.75 (−1.4)	0.24 (0.3)
(4) + volume, volatility, lagged taker	0.00 (0.0)	−0.82 (−1.7)	0.08 (0.1)
(5) 1m BVC alone	0.33 (0.9)	−0.23 (−0.4)	0.24 (0.3)
(6) taker vs 1m BVC: taker	−0.12 (−0.4)	−0.59 (−1.1)	0.60 (0.7)
(6) taker vs 1m BVC: 1m BVC	0.40 (0.9)	0.10 (0.2)	−0.09 (−0.1)

Coefficient (basis points per standard deviation) of observed taker imbalance, or of one-minute BVC imbalance where indicated, in regressions of the k-hour return on standardised regressors; Newey-West t-statistics in parentheses. (2) adds the standardised bar return; (3) adds three lagged hourly returns and 24-hour momentum; (4) adds standardised volume, Garman-Klass volatility and the previous hour's imbalance; (5) uses one-minute BVC imbalance alone; (6) includes observed and one-minute BVC imbalance together.

7 Out-of-sample forecasting and economic value

7.1 Design

Target and walk-forward. The target is the four-hour log return $y_t = \log C_{t+4} - \log C_t$, formed at the close of hour t. The first six months after the feature warm-up are used only for training. The remaining sample (2022-07-10 to 2026-10-31, 37,593 hours) is cut into 53 consecutive 30-day test blocks. Before each block every model is re-estimated on all observations whose label is observable at the block start, i.e. with $t + 4$ strictly before the block (purging); nothing from the block enters estimation. Within each training window the last 20% serves as a validation block, itself purged from the inner training rows, on which hyper-parameters are chosen before the model is refitted on the full window. Winsorisation bounds (1st and 99th percentiles) and standardisation moments are estimated on the training window only. Automated tests replace every bar after t by noise and verify that features at t are unchanged, and check the purge condition for every fold (Appendix B).

Models. Following Gu et al. (2020), models forecast the return relative to zero. The elastic net is fitted without intercept on standardised features, with the penalty chosen from a 13-point grid and the ℓ_1 share from $\{0.1, 0.5, 0.9\}$ on the validation block. As a non-linear benchmark we use gradient-boosted trees with depth three, at least 300 observations per leaf and a learning rate of

Table 5: Out-of-sample forecast accuracy, four-hour BTC/USDT returns

Model	R^2_{OOS} vs. 0 (%)	R^2_{OOS} vs. mean (%)	CW t vs. 0	IC	95% CI	Hit rate	Hit t
Historical mean	−0.075	0.000	–	−0.002	[−0.021, 0.015]	–	–
EN: TI	−0.055	0.020	1.40*	0.002	[−0.014, 0.019]	0.491	−2.14
EN: MS_{proxy}	−0.139	−0.064	−0.61	−0.005	[−0.024, 0.013]	0.488	−2.05
EN: MS	−0.006	0.070	0.86	−0.002	[−0.019, 0.014]	0.494	−0.92
EN: ALL	−0.025	0.050	1.34*	0.019	[0.004, 0.035]	0.508	1.44
EN: Flow_{BVC}	−0.081	−0.005	−1.35	−0.021	[−0.038, −0.006]	0.475	−3.99
EN: $\text{Flow}_{\text{taker}}$	−0.036	0.040	−0.41	−0.007	[−0.021, 0.009]	0.492	−1.26
GBM: MS	−0.284	−0.208	0.39	0.000	[−0.018, 0.019]	0.498	−0.66
GBM: ALL	−0.262	−0.187	0.98	0.003	[−0.014, 0.020]	0.494	−1.72

Monthly expanding-window walk-forward, 2022-07-10 to 2026-10-31 (37,593 hourly forecasts). EN: elastic net; GBM: gradient-boosted trees. CW: Clark-West statistic against the zero forecast (one-sided). IC: Spearman correlation of forecast and outcome, with 95% block-bootstrap interval. Hit t: HAC t-statistic of the hit rate against 50%. *, **, ***: 10%, 5%, 1% significance.

0.03, with the number of rounds chosen on the validation block. We also report the historical-mean forecast. We estimate every model on every feature set defined in Section 4 and report all of them.

Statistical evaluation. The out-of-sample R^2 relative to a benchmark b is $R^2_{OOS} = 1 - \sum(y - \hat{y})^2 / \sum(y - \hat{y}^b)^2$ (Campbell and Thompson, 2008). Because the zero forecast and smaller feature sets are nested in larger ones, we test equal predictive accuracy with the Clark and West (2007) statistic (one-sided), which adjusts for the noise that estimating irrelevant parameters adds to the larger model, and report the Diebold and Mariano (1995) statistic, which compares realised accuracy. Overlapping four-hour targets make forecast errors serially correlated, so all t-statistics use Newey and West (1987) standard errors with $\max(24, 2h)$ lags. The information coefficient (rank correlation of forecast and outcome) is reported with a 95% moving-block bootstrap interval with 48-hour blocks (Künsch, 1989).

Economic evaluation. Each hour a signal $s_t \in \{-1, 0, 1\}$ is formed and held for four hours with a quarter of capital per sleeve (Jegadeesh and Titman, 1993), so the position over $(t, t+1]$ is $\frac{1}{4}\sum_{k=0}^{3} s_{t-k}$; every forecast is used and no exposure is double counted. The *sign* rule sets $s_t = \text{sign}(\hat{y}_t)$, which is zero when the model predicts exactly zero; the *selective* rule trades only when $|\hat{y}_t|$ exceeds the 80th percentile of the model's absolute in-sample forecasts in the current training window. Costs of $c \in \{0, 2, 5, 10\}$ basis points are charged per unit change in position. We report the break-even cost at which net profit is zero, the HAC t-statistic of the intercept in a regression of strategy returns on BTC returns, the number of calendar years with a positive Sharpe ratio, and the deflated Sharpe ratio (DSR) of Bailey and López de Prado (2014). The DSR is the probability that a strategy's true Sharpe ratio exceeds the maximum expected among all 16 model and rule combinations under the null of no skill, given their dispersion, the sample length and the non-normality of returns.

7.2 Forecast accuracy

Table 5 reports the main results for BTC/USDT. All 8 models have a negative out-of-sample R^2 against the zero forecast; the best is EN: MS at −0.006%. The historical mean is also worse than

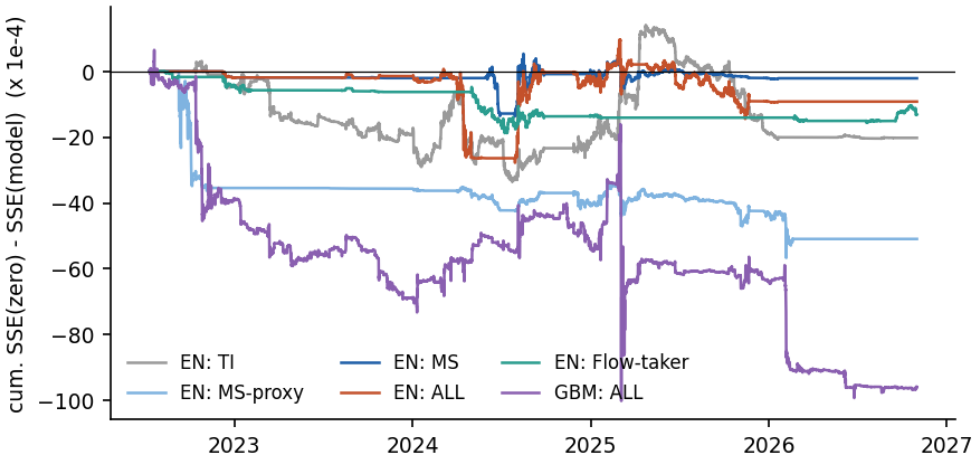


Figure 4: Cumulative squared-error difference between the zero forecast and each model, BTC/USDT four-hour returns. A rising line means the model is more accurate than the zero forecast.

Table 6: Incremental predictive information, BTC/USDT

Model	Benchmark	Nested	R^2_{OOS} rel. (%)	CW t	CW p	DM t	DM p
EN: ALL	EN: TI	yes	0.030	2.41***	0.008	0.33	0.741
EN: MS	EN: TI	yes	0.050	2.85***	0.002	0.55	0.585
EN: MS	EN: MS_{proxy}	yes	0.133	2.59***	0.005	1.53	0.125
EN: MS_{proxy}	EN: TI	yes	−0.084	1.49*	0.068	−0.71	0.476
GBM: ALL	EN: TI	yes	−0.207	1.73**	0.042	−0.88	0.378
EN: $\text{Flow}_{\text{taker}}$	EN: Flow_{BVC}	no	0.045	–	–	0.97	0.332

R^2_{OOS} rel.: out-of-sample R^2 of the model relative to the benchmark. CW: Clark-West test of the null that the larger model adds nothing to the nested benchmark (one-sided). DM: Diebold-Mariano test of equal mean squared error (two-sided). The observed-flow and BVC-flow models are not nested, so only DM is reported.

zero: trailing average returns are a noisy estimate of the drift, and in this sample the early training windows are dominated by the 2022 bear market, which the following recovery reversed (Campbell and Thompson, 2008). Clark-West statistics against zero are significant at the 10% level only for the technical-indicator model ($p = 0.080$) and the full model ($p = 0.091$). The information coefficients are all within about two hundredths of zero. The confidence interval excludes zero for the full model (IC = 0.019) and, with the wrong sign, for the BVC-flow model (IC = −0.021), whose coefficients are unstable across folds. The observed-flow model, which uses the matched constructions from exchange-reported taker volume, does no better than its BVC counterpart.

Figure 4 plots the cumulative difference in squared errors between the zero forecast and each model (Welch and Goyal, 2008). No line trends upwards for an extended period; gains in some months are offset by losses in others.

7.3 Incremental information

Table 6 asks whether each richer feature set improves on a simpler one. The Clark-West test rejects the null that microstructure variables add nothing to technical indicators ($p = 0.002$ for MS over TI

Table 7: Single-predictor out-of-sample test

						R^2_{OOS} (CW p) by half	
Predictor	Horizon	R^2_{OOS} (%)	CW t	CW p	IC	first	second
BTC/USDT							
Observed taker imbalance	1h	0.012	3.19***	0.0007	0.048	0.009 (0.032)	0.016 (0.005)
Observed taker imbalance	2h	0.021	3.80***	0.0001	0.044	0.007 (0.022)	0.038 (0.001)
Observed taker imbalance	4h	0.016	2.62***	0.0044	0.033	0.003 (0.233)	0.032 (0.006)
BVC imbalance (1h bars)	1h	−0.001	0.74	0.2286	0.030	0.007 (0.183)	−0.011 (0.370)
BVC imbalance (1h bars)	2h	−0.007	−0.56	0.7119	0.015	−0.010 (0.848)	−0.004 (0.489)
BVC imbalance (1h bars)	4h	−0.008	−0.35	0.6353	−0.034	−0.007 (0.484)	−0.010 (0.836)
BVC imbalance (1m bars)	1h	0.014	2.39***	0.0085	0.060	0.027 (0.015)	0.000 (0.111)
BVC imbalance (1m bars)	2h	0.017	2.55***	0.0053	0.054	0.023 (0.020)	0.009 (0.062)
BVC imbalance (1m bars)	4h	0.002	1.04	0.1498	0.038	−0.001 (0.319)	0.005 (0.170)
ETH/USDT							
Observed taker imbalance	1h	−0.013	0.63	0.2652	−0.035	−0.025 (0.227)	−0.003 (0.911)
Observed taker imbalance	2h	−0.006	0.14	0.4443	0.002	−0.018 (0.616)	0.004 (0.198)
Observed taker imbalance	4h	−0.015	−0.69	0.7560	−0.016	−0.031 (0.772)	−0.002 (0.499)
BVC imbalance (1h bars)	1h	−0.006	0.42	0.3383	−0.040	−0.023 (0.448)	0.008 (0.127)
BVC imbalance (1h bars)	2h	−0.004	0.41	0.3405	−0.043	−0.005 (0.358)	−0.004 (0.423)
BVC imbalance (1h bars)	4h	0.009	1.20	0.1151	−0.026	−0.011 (0.620)	0.025 (0.063)
BVC imbalance (1m bars)	1h	−0.005	−0.12	0.5462	−0.018	−0.011 (0.612)	0.001 (0.373)
BVC imbalance (1m bars)	2h	−0.005	−0.54	0.7052	0.023	−0.007 (0.733)	−0.003 (0.562)
BVC imbalance (1m bars)	4h	−0.006	−1.19	0.8829	0.013	−0.010 (0.850)	−0.004 (0.725)

One-variable expanding-window regression, re-estimated monthly with purged labels; the coefficient is learned from past data only. R^2_{OOS} against the zero forecast; CW: Clark-West test against zero (one-sided); IC: Spearman correlation of forecast and outcome. The last two columns split the out-of-sample period into halves.

and $p = 0.008$ for ALL over TI) and that observed flow and intrahour measures add nothing to the OHLCV proxies ($p = 0.005$). In each case, however, the realised gain in mean squared error is a few hundredths of a percent and the Diebold-Mariano test does not reject equal accuracy. This is the pattern Clark-West is designed to detect: the larger model's population coefficients are not all zero, but the information they carry is smaller than the noise introduced by estimating them, so the larger model does not forecast better in practice.

The model selection inside each training window tells the same story. For the elastic net the validation step chooses the penalty that sets every coefficient to zero in 57% of the 53 folds with the OHLCV proxies, 68% with observed flow and intrahour measures, and 58% with all features, against 25% with technical indicators alone. Most of the time, the data prefer no forecast at all.

7.4 A single predictor: observed flow against its proxy

The flow models above combine five constructions and tune a penalty on a short validation block, which may discard a weak but genuine signal. Section 6 suggests a direct test: does the current hour's observed taker imbalance, on its own, forecast returns out of sample, and does its BVC counterpart? For each predictor we estimate a one-variable regression without intercept on the standardised predictor in the same expanding monthly walk-forward, with purged labels and moments from the training window only, and no tuning.

Table 7 gives a clean contrast for BTC. Observed taker imbalance has a positive out-of-sample

Table 8: Trading on the single observed-flow forecast, two-hour holding period

		Ann. ret. (%)	Sharpe at cost (bps)			Turnover	Break-even
Asset	Rule	gross	0	1	2	per day	(bps)
BTC/USDT	sign	30.0	0.87	−0.34	−1.54	11.4	0.72
BTC/USDT	selective	26.4	1.43	0.23	−0.97	6.1	1.19
ETH/USDT	sign	10.6	0.23	−0.67	−1.56	11.3	0.26
ETH/USDT	selective	5.9	0.26	−0.62	−1.50	5.4	0.30

Overlapping-sleeve rule on the out-of-sample forecast from observed taker imbalance alone (Table 7, two-hour horizon). Sign: trade every hour in the direction of the forecast; selective: only when $|\hat{y}_t|$ exceeds the 80th percentile of in-sample absolute forecasts. Break-even: cost per unit turnover at which net profit is zero.

R^2 at every horizon, 0.021% at two hours with a Clark-West p-value of 0.0001 and an information coefficient of 0.044. The result holds in both halves of the test period (0.007% with $p = 0.022$ in the first and 0.038% with $p = 0.0005$ in the second). Hourly BVC imbalance has a negative out-of-sample R^2 at every horizon and no significant Clark-West statistic. BVC on one-minute bars lies in between, with an out-of-sample R^2 of 0.017% at two hours ($p = 0.005$), in line with the in-sample gradient of Section 6. Because the coefficient is re-estimated from past data each month, the sign of the relation is learned in real time; the predictor itself, however, was suggested by the full-sample analysis, so the test is not pre-registered. Even a Bonferroni correction for all 45 candidate features leaves the two-hour result for observed flow significant at the 1% level. For ETH none of the three predictors forecasts returns, consistent with the weaker in-sample response.

Two lessons follow. First, the information that order flow carries about short-horizon BTC returns is detectable out of sample when flow is measured, partly recoverable from one-minute prices, and absent from hourly bars. Second, the multivariate elastic net that includes the same variable fails to exploit it: with a weak signal and many correlated candidates, cross-validated shrinkage on a short validation block usually selects the empty model. The signal is nonetheless far too small to trade. Table 8 applies the overlapping-sleeve rule to the observed-flow forecast with a two-hour holding period. Gross Sharpe ratios are 0.87 for the sign rule and 1.43 for the selective rule, but the rules turn over their position many times a day, and their break-even costs are 0.7 and 1.2 basis points per unit of turnover. That is an order of magnitude below the standard fee of ten basis points, and below the taker fees of high-volume accounts, which remain at a few basis points; a strategy that must trade immediately on an hourly signal cannot rely on maker rebates. The predictable component is a statistical regularity, not an exploitable strategy.

7.5 Economic value

Table 9 translates the forecasts into trading rules. Before costs, most rules have Sharpe ratios between −1 and 1, and only 1 of the 16 combinations keeps a positive Sharpe ratio at five basis points. One combination stands out: the sign rule applied to the full elastic net, with a gross Sharpe ratio of 1.24, a break-even cost of 9.0 basis points and almost no market exposure (beta −0.01, alpha $t = 2.7$). Three diagnostics argue against reading it as evidence of a profitable signal. Its probabilistic Sharpe ratio against zero is 0.995, but once we account for having selected it from 16 candidates, the deflated Sharpe ratio is 0.70, below conventional thresholds. Its Sharpe ratio ranges

Table 9: Trading performance of forecast-based rules, BTC/USDT

	Ann. ret. (%)	Sharpe at cost (bps)			Turnover	Break-even	α t	Years	DSR
Model	gross	0	2	5	per day	(bps)	(HAC)	$SR > 0$	
Sign rule									
EN: TI	7.8	0.39	−0.19	−1.05	2.61	1.4	0.78	2/5	0.11
EN: MS_{proxy}	−14.1	−0.38	−0.59	−0.90	0.81	−3.7	−0.88	0/5	0.00
EN: MS	−10.9	−0.40	−0.64	−1.00	0.75	−3.3	−0.83	1/5	0.00
EN: ALL	35.5	1.24	0.97	0.55	1.04	9.0	2.68	3/5	0.70
EN: $Flow_{BVC}$	−22.8	−0.84	−1.00	−1.24	0.59	−10.4	−1.54	0/5	0.00
EN: $Flow_{taker}$	−5.5	−0.17	−0.42	−0.80	0.72	−1.3	−0.17	3/5	0.01
GBM: MS	−10.1	−0.05	−0.46	−1.07	2.33	−0.2	0.25	2/5	0.02
GBM: ALL	−10.7	−0.06	−0.38	−0.87	1.85	−0.4	0.46	3/5	0.02
Selective rule (top quintile of |forecast|)									
EN: TI	10.9	0.71	0.34	−0.21	0.83	3.9	1.51	4/5	0.28
EN: MS_{proxy}	−7.8	−0.47	−0.66	−0.93	0.38	−5.1	−1.21	1/5	0.00
EN: MS	0.6	0.11	−0.12	−0.46	0.33	1.0	0.13	3/5	0.03
EN: ALL	6.1	0.47	0.26	−0.05	0.43	4.5	0.96	4/5	0.14
EN: $Flow_{BVC}$	−8.4	−0.59	−0.75	−0.98	0.29	−7.5	−1.02	1/5	0.00
EN: $Flow_{taker}$	−4.9	−0.45	−0.74	−1.17	0.39	−3.2	−0.89	1/5	0.00
GBM: MS	8.9	0.49	0.26	−0.08	0.71	4.3	1.03	2/5	0.15
GBM: ALL	−2.4	0.00	−0.26	−0.65	0.76	0.0	0.14	2/5	0.02
Buy and hold	34.8	0.87							

Overlapping-sleeve implementation with a four-hour holding period. Costs are per unit change in position. Break-even: cost per unit turnover at which net profit is zero. α t: HAC t-statistic of the intercept in a regression of gross strategy returns on BTC returns. Years $SR > 0$: calendar years with positive gross Sharpe ratio. DSR: deflated Sharpe ratio (Bailey and López de Prado, 2014) across all 16 model and rule combinations.

from −1.8 to 1.9 across calendar years and is positive in 3/5 of them. And it is not the model with the best forecast accuracy, so its performance does not reflect a systematic forecasting advantage. No rule has a break-even cost above 9.0 basis points, so a trader paying the standard fee of ten basis points per side would have lost money with every rule.

8 Robustness

Second asset. We repeat the full analysis for ETH/USDT (Appendix Tables 23, 20 and 24). The measurement results are almost identical. Hourly BVC imbalance has a correlation of 0.49 with observed imbalance, the in-sample information ceiling of return-based classifiers is 0.24 on hourly and 0.29 on one-minute bars, Garman-Klass volatility has a correlation of 0.99 with realised volatility, and the Corwin-Schultz estimate has a median of 20.7 basis points against a tick of 0.043. The zero-fee promotion offers a useful comparison here. On BTC/USDT, which was included, the hourly BVC correlation rises from 0.51 under standard fees to 0.60 during the promotion; on ETH/USDT, which was not, it is 0.49 and 0.50. The fee change altered the relation between prices and aggressor flow only on the asset it applied to.

The forecasting results for ETH differ in one respect. The technical-indicator model is the only one with a positive out-of-sample R^2 (0.102%, Clark-West $p = 0.001$), and adding microstructure variables makes forecasts significantly *worse*: the Diebold-Mariano p-value for ALL against TI is

Table 10: Stability of the return response to observed flow

		Observed imbalance		Observed imbalance | z_t	
Sample	Hours	$k = 2$	$k = 4$	$k = 2$	$k = 4$
BTC/USDT					
Full sample	41,977	−1.30 (−4.0)	−1.33 (−2.6)	−1.41 (−3.7)	−1.55 (−2.6)
Standard fees	35,809	−1.37 (−3.9)	−1.40 (−2.5)	−1.52 (−3.7)	−1.59 (−2.5)
Zero-fee window	6,168	−1.52 (−1.7)	−2.06 (−1.4)	−1.86 (−1.7)	−3.26 (−1.7)
Jan 2022 - Jun 2024	21,506	−1.49 (−2.9)	−0.98 (−1.2)	−1.72 (−2.9)	−1.36 (−1.5)
Jul 2024 - Oct 2026	20,471	−1.27 (−2.9)	−1.63 (−2.4)	−1.33 (−2.6)	−1.69 (−2.1)
ETH/USDT					
Full sample	41,977	−0.53 (−1.1)	0.55 (0.8)	−0.79 (−1.5)	0.21 (0.3)
Standard fees	35,809	−0.67 (−1.3)	0.68 (0.9)	−0.88 (−1.6)	0.30 (0.3)
Zero-fee window	6,168	0.46 (0.4)	−0.70 (−0.4)	−0.07 (−0.1)	−0.96 (−0.5)
Jan 2022 - Jun 2024	21,506	−0.44 (−0.7)	0.10 (0.1)	−0.64 (−0.9)	0.23 (0.2)
Jul 2024 - Oct 2026	20,471	−0.63 (−0.9)	0.87 (0.8)	−0.97 (−1.3)	0.03 (0.0)

Coefficients (basis points per standard deviation) and Newey-West t-statistics from regression (1) at horizons of two and four hours, by sub-sample. The last two columns include the standardised bar return z_t as a control.

0.004, with ALL less accurate. No microstructure or flow model beats the zero forecast. The best trading rule is again one that uses no microstructure information, the sign rule on EN: TI with a gross Sharpe ratio of 1.12, and its deflated Sharpe ratio of 0.75 is below conventional thresholds. The reversal after aggressor flow found for BTC does not replicate for ETH at conventional significance levels: the two-hour coefficient is negative in the full sample, under standard fees and in both halves, but about half as large as for BTC, with t-statistics between −0.7 and −1.7, and it is positive during the zero-fee window (Table 10). In ETH, BVC imbalance has a positive coefficient at eight to twelve hours ($t \approx 2.3$, Appendix Table 22); the bar-return coefficient is similar, so this reflects return continuation rather than information in flow.

Stability of the flow response. Table 10 re-estimates regression (1) in sub-samples. For BTC the two-hour reversal is of similar size under standard fees, during the zero-fee window and in both halves of the sample, and remains when the bar return is held fixed. It is therefore not an artefact of the promotion or of one market phase.

Forecast horizon. Table 11 repeats the walk-forward for horizons of 1, 8 and 24 hours. The pattern matches ETH: the technical-indicator model has small positive out-of-sample R^2 at one, eight and 24 hours, the largest being 0.109% at eight hours (Clark-West $p = 0.002$), while the microstructure feature sets never improve on it.

Estimation window. Replacing the expanding window by a six-month rolling window, which adapts faster to changing conditions at the cost of noisier estimates, makes every model worse, with out-of-sample R^2 of about −0.4% and no significant Clark-West statistic (Appendix Table 16). Whatever predictability exists is not a short-lived, local relation that a shorter window would

Table 11: Forecast horizon, BTC/USDT

Horizon	EN: TI	EN: MS_{proxy}	EN: MS	EN: ALL
1h	0.018 (0.01)	−0.025 (0.73)	0.000 (0.06)	−0.004 (0.05)
4h	−0.055 (0.08)	−0.139 (0.73)	−0.006 (0.19)	−0.025 (0.09)
8h	0.109 (0.00)	0.010 (0.15)	0.007 (0.27)	0.034 (0.04)
24h	0.072 (0.03)	−0.572 (0.78)	−0.841 (0.90)	−0.108 (0.32)

Out-of-sample R^2 against the zero forecast (%), Clark-West p-value against zero in parentheses.

capture.

Feature groups. Dropping one feature group at a time from the full elastic net (Appendix Table 17) leaves the out-of-sample R^2 between about −0.17% and +0.02%. The only positive value arises when the volatility group is removed, and it is too small to matter; no single group is responsible for the negative results.

Conditional predictability. Predictability could be concentrated in particular states, for example when volatility is high or liquidity is low. Appendix Table 18 splits the out-of-sample period by causal volatility and illiquidity terciles, trading session, weekday and fee regime. Out-of-sample R^2 is close to zero in every cell and no split shows consistent gains across models.

9 Discussion and limitations

What the results do and do not show. The forecasting results are negative, but they are informative because the design rules out the usual explanations. The features are not too few or too coarse: observed flow, trade counts and one-minute realised measures are included. The models are not too simple: shallow boosted trees do no better than the elastic net. The evaluation is not too short: it covers 53 months and a full cycle of bear, recovery and bull markets. What the results show is that, at hourly sampling and a four-hour horizon, the predictable component of BTC and ETH returns linked to these variables is a fraction of a basis point per hour, an order of magnitude below the cost of trading on it.

Limitations. We do not observe the limit order book, so we cannot measure quoted or effective spreads directly or compute order-flow imbalance at the best quotes (Cont et al., 2014), which is known to have predictive content at horizons of seconds to minutes (Kolm et al., 2023). Our results therefore speak to bar-based and aggregated trade data, not to high-frequency order-book signals. We study one exchange; aggressor flow on Binance is a large but partial view of a fragmented market (Makarov and Schoar, 2020), and cross-venue flow may be more informative. Taker flags identify the aggressor, not the informed party, and in markets with sophisticated liquidity providers the passive side may be better informed, which is one reading of the reversal in Section 6. Finally, the feature sets, models, walk-forward design and trading rules were fixed before estimation, and we report every specification we ran. The flow-response regressions of Section 6 and the single-predictor test of Section 7 were added after the first results pointed to observed imbalance. The former are

descriptive; the latter, although estimated strictly out of sample, should be read with this sequence and the number of candidate features in mind.

10 Conclusion

Exchange-reported aggressor volume lets us check what bar-based microstructure variables measure. In hourly Bitcoin and Ether data they measure volatility very well and order flow poorly, and the part of the flow they miss is precisely the part that, in Bitcoin, carries information about future prices. That information is real but small: it implies a reversal of one to two basis points over a few hours, which a single-variable model detects out of sample but multivariate models discard, and which does not survive transaction costs or a correction for the number of strategies tried. For researchers, the implication is that BVC-based imbalance, VPIN and Kyle's lambda computed from bars should not be interpreted as measures of order flow when the aggressor side is available; that, when it is not, BVC should be computed on the finest bars available and recalibrated as its scale drifts; and that high-low spread estimators on liquid crypto pairs measure volatility. For forecasting, the binding constraint at this horizon is not model complexity but the size of the signal relative to costs. Order-book data at higher frequency, and flow measured across venues, are the natural next steps.

Data and code availability. All data are public and are downloaded by the scripts in the replication package, which also contains the leakage tests and reproduces every table and figure: https://github.com/a0x0h/what-bars-cannot-see.

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References

Farshid Abdi and Angelo Ranaldo. A simple estimation of bid-ask spreads from daily close, high, and low prices. *Review of Financial Studies*, 30(12):4437–4480, 2017.

Saketh Aleti and Bruce Mizrach. Bitcoin spot and futures market microstructure. *Journal of Futures Markets*, 41(2):194–225, 2021.

Diego Amaya, Peter Christoffersen, Kris Jacobs, and Aurelio Vasquez. Does realized skewness predict the cross-section of equity returns? *Journal of Financial Economics*, 118(1):135–167, 2015.

Yakov Amihud. Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets*, 5(1):31–56, 2002.

Torben G. Andersen and Oleg Bondarenko. VPIN and the flash crash. *Journal of Financial Markets*, 17:1–46, 2014.

David H. Bailey and Marcos López de Prado. The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5):94–107, 2014.

Ole E. Barndorff-Nielsen and Neil Shephard. Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics*, 2(1):1–37, 2004.

Jean-Philippe Bouchaud, Julius Bonart, Jonathan Donier, and Martin Gould. *Trades, Quotes and Prices: Financial Markets Under the Microscope*. Cambridge University Press, 2018.

Alexander Brauneis, Roland Mestel, Ryan Riordan, and Erik Theissen. How to measure the liquidity of cryptocurrency markets? *Journal of Banking & Finance*, 124:106041, 2021.

John Y. Campbell and Samuel B. Thompson. Predicting excess stock returns out of sample: Can anything beat the historical average? *Review of Financial Studies*, 21(4):1509–1531, 2008.

Bidisha Chakrabarty, Roberto Pascual, and Andriy Shkilko. Evaluating trade classification algorithms: Bulk volume classification versus the tick rule and the Lee-Ready algorithm. *Journal of Financial Markets*, 25:52–79, 2015.

Tarun Chordia, Richard Roll, and Avanidhar Subrahmanyam. Order imbalance, liquidity, and market returns. *Journal of Financial Economics*, 65(1):111–130, 2002.

Todd E. Clark and Kenneth D. West. Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1):291–311, 2007.

Lin William Cong, Xi Li, Ke Tang, and Yang Yang. Crypto wash trading. *Management Science*, 69 (11):6427–6454, 2023.

Rama Cont, Arseniy Kukanov, and Sasha Stoikov. The price impact of order book events. *Journal of Financial Econometrics*, 12(1):47–88, 2014.

Shane A. Corwin and Paul Schultz. A simple way to estimate bid-ask spreads from daily high and low prices. *Journal of Finance*, 67(2):719–760, 2012.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.

David Easley, Marcos López de Prado, and Maureen O'Hara. Flow toxicity and liquidity in a high-frequency world. *Review of Financial Studies*, 25(5):1457–1493, 2012.

David Easley, Marcos López de Prado, and Maureen O'Hara. Discerning information from trade data. *Journal of Financial Economics*, 120(2):269–285, 2016.

David Easley, Marcos López de Prado, Maureen O'Hara, and Zhibai Zhang. Microstructure in the machine age. *Review of Financial Studies*, 34(7):3316–3363, 2021.

David Easley, Maureen O'Hara, Songshan Yang, and Zhibai Zhang. Microstructure and market dynamics in crypto markets. Working paper, SSRN 4814346, 2024.

Mark B. Garman and Michael J. Klass. On the estimation of security price volatilities from historical data. *Journal of Business*, 53(1):67–78, 1980.

Lawrence R. Glosten and Paul R. Milgrom. Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1):71–100, 1985.

Ruslan Y. Goyenko, Craig W. Holden, and Charles A. Trzcinka. Do liquidity measures measure liquidity? *Journal of Financial Economics*, 92(2):153–181, 2009.

Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5):2223–2273, 2020.

Campbell R. Harvey, Yan Liu, and Heqing Zhu. . . . and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68, 2016.

Mohammad R. Jahan-Parvar and Filip Zikes. When do low-frequency measures really measure effective spreads? Evidence from equity and foreign exchange markets. *Review of Financial Studies*, 36(10):4190–4232, 2023.

Narasimhan Jegadeesh and Sheridan Titman. Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1):65–91, 1993.

Petter N. Kolm, Jeremy Turiel, and Nicholas Westray. Deep order flow imbalance: Extracting alpha at multiple horizons from the limit order book. *Mathematical Finance*, 33(4):1044–1081, 2023.

Hans R. Künsch. The jackknife and the bootstrap for general stationary observations. *Annals of Statistics*, 17(3):1217–1241, 1989.

Albert S. Kyle. Continuous auctions and insider trading. *Econometrica*, 53(6):1315–1335, 1985.

Charles M. C. Lee and Mark J. Ready. Inferring trade direction from intraday data. *Journal of Finance*, 46(2):733–746, 1991.

Yukun Liu and Aleh Tsyvinski. Risks and returns of cryptocurrency. *Review of Financial Studies*, 34(6):2689–2727, 2021.

Igor Makarov and Antoinette Schoar. Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics*, 135(2):293–319, 2020.

Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.

Marios A. Panayides, Thomas D. Shohfi, and Jared D. Smith. Bulk volume classification and information detection. *Journal of Banking & Finance*, 103:113–129, 2019.

L. C. G. Rogers and S. E. Satchell. Estimating variance from high, low and closing prices. *Annals of Applied Probability*, 1(4):504–512, 1991.

Richard Roll. A simple implicit measure of the effective bid-ask spread in an efficient market. *Journal of Finance*, 39(4):1127–1139, 1984.

Ivo Welch and Amit Goyal. A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 21(4):1455–1508, 2008.

Dennis Yang and Qiang Zhang. Drift-independent volatility estimation based on high, low, open, and close prices. *Journal of Business*, 73(3):477–491, 2000.

A Feature definitions

Table 12: Feature definitions

Feature	Set	Definition (bar t; r log return, V volume, B taker-buy volume, C, H, L, O prices)
Spread and price impact		
cs_spread_24	MS_{proxy}	24h mean of the two-bar Corwin and Schultz (2012) estimator
ar_spread_24	MS_{proxy}	$\sqrt{\max(0, \overline{4(c_{t-1}-\eta_{t-1})(c_{t-1}-\eta_t)})}$ over 24h, $\eta = (\log H + \log L)/2$ (Abdi and Ranaldo, 2017)
roll_spread_48	MS_{proxy}	$2\sqrt{\max(0, -\operatorname{cov}(r_t, r_{t-1}))}$ over 48h (Roll, 1984)
amihud_24	MS_{proxy}	log of 24h mean of $\lvert r_t\rvert/(C_t V_t)$ (Amihud, 2002)
sqrt_impact_24	MS_{proxy}	log of 24h mean of $\lvert r_t\rvert/\sqrt{V_t/\bar{V}_{168}}$
Order flow, BVC proxy (signed volume $S_t = (2\Phi(r_t/\hat{\sigma}_{168}) - 1)V_t$)		
ofi_1, ofi_24	MS_{proxy}, Flow_{BVC}	$S_t/\bar{V}_{168}$; $\sum_{24} S/\sum_{24} V$
ofi_accel	MS_{proxy}, Flow_{BVC}	6h imbalance minus 24h imbalance
vpin_24	MS_{proxy}, Flow_{BVC}	$\sum_{24} \lvert S\rvert/\sum_{24} V$
kyle_lambda_48	MS_{proxy}, Flow_{BVC}	$\operatorname{cov}(r, SC)/\operatorname{var}(SC)$ over 48h
Order flow, observed (signed volume $2B_t - V_t$) and trade activity		
taker_imb_1, _24, _accel, taker_vpin_24, kyle_lambda_true_48	MS, $\text{Flow}_{\text{taker}}$	as the BVC versions, with observed signed volume
taker_imb_late15	MS	taker imbalance in minutes 45–59 of the hour
trades_z_168, trade_size_z_168	MS	168h z-scores of log trade count and log average dollar trade size
Volatility		
gk_vol_24, rs_vol_24, yz_vol_24	MS_{proxy}	log 24h Garman and Klass (1980), Rogers and Satchell (1991), Yang and Zhang (2000) volatility
vol_ratio_24_168	MS_{proxy}	log ratio of 24h to 168h Garman-Klass volatility
vol_of_vol_168	MS_{proxy}	168h standard deviation of log 24h Garman-Klass volatility
rv_1m_1, rv_1m_24	MS	log realised variance from 1-minute returns, current hour and 24h
jump_share_24	MS	24h mean of $\max(RV - BV, 0)/RV$ (Barndorff-Nielsen and Shephard, 2004)
rskew_24	MS	24h mean of hourly realised skewness (Amaya et al., 2015)
Efficiency, bar structure, activity		
eff_ratio_24	MS_{proxy}	$\lvert\log C_t - \log C_{t-24}\rvert/\sum_{24} \lvert r\rvert$
autocorr_48	MS_{proxy}	48h first-order autocorrelation of r
var_ratio_4_168	MS_{proxy}	variance of 4h returns over 4 times variance of 1h returns, 168h
body_share, upper_wick, lower_wick (24h)	MS_{proxy}	candle body and wicks as share of the high-low range
volume_z_168, volume_trend_24_168	MS_{proxy}	z-score of log volume; log ratio of 24h to 168h mean volume
Technical controls		
mom_1, _4, _24, _168	TI	$\log C_t - \log C_{t-k}$
rsi_14, macd_hist, bb_pctb_20	TI	RSI(14); MACD(12,26,9) histogram over price; Bollinger %B(20,2)
ma_gap_24, ma_gap_168, atr_14	TI	log gap to moving average; ATR(14) over price

B Leakage controls and a positive control

The replication package contains automated tests that run in a few seconds. (i) *Causality of features*: for three cut-off dates, every bar after the cut-off is multiplied by random noise and every feature at or before the cut-off is recomputed; the test fails if any value changes. The same test is applied to the intrahour features built from one-minute bars. (ii) *Target definition*: y_t must equal $\log C_{t+h} - \log C_t$ exactly. (iii) *Purging*: for every fold and every horizon, the last training label must be observed before the first test bar, and the same must hold for the inner validation split. (iv) *Analyses*: every validation and robustness function is run on synthetic data and checked for internal consistency (for example, the in-sample ceiling must weakly exceed the squared correlation of BVC), a pure-noise predictor must not produce an out-of-sample gain in the single-predictor walk-forward, and a planted flow signal must be recovered from observed flow but not from technical momentum.

As a positive control we run the full pipeline on a synthetic two-year hourly series with GARCH-type volatility, in which a persistent latent order flow drives the taker share of volume and each hour's return loads negatively on the average flow of the previous four hours. The contemporaneous link between flow and the bar's return is deliberately weak, so BVC recovers the flow only poorly, as in the real data. The pipeline should find the signal with observed-flow features and much less of it with BVC features, and nothing with technical indicators. It does: the observed-flow elastic net reaches an out-of-sample R^2 of about 3% with a Clark-West t-statistic above 10, the BVC-flow model about zero, and the technical model a negative R^2. A pipeline that leaked future information would instead show gains for every feature set. The script `scripts/make_synthetic.py` generates the series.

C Additional tables

Table 13: Alternative flow classifiers vs. observed taker imbalance, BTC/USDT

	Pearson		Same sign	
Classifier	1h	24h	1h	24h
BVC Normal, sigma 24h (1h bars)	0.503	0.516	0.716	0.695
BVC Normal, sigma 168h (1h bars)	0.489	0.507	0.716	0.693
BVC Normal, sigma 720h (1h bars)	0.484	0.505	0.716	0.691
BVC Student-t(3), sigma 168h (1h bars)	0.494	0.507	0.716	0.687
Tick rule (1h bars)	0.430	0.456	0.716	0.666
BVC Normal, sigma 1440m (1m bars)	0.603	0.602	0.748	0.725
Tick rule (1m bars)	0.597	0.617	0.728	0.731

Correlation and sign agreement between proxy imbalance and observed taker imbalance, per hour (1h) and over rolling 24-hour sums (24h). Student-$t(3)$ is rescaled to unit variance.

Table 14: Information ceiling by period

Period	Hours	Ceiling	corr^2(BVC)	Slope	SD imb.	Ceiling (1m)
BTC/USDT						
full sample	42,315	0.247	0.239	0.128	0.130	0.338
zero-fee window	6,168	0.374	0.360	0.038	0.030	0.344
standard fees	36,147	0.267	0.257	0.142	0.140	0.362
2022	8,718	0.242	0.234	0.065	0.066	0.248
2023	8,757	0.278	0.253	0.106	0.099	0.370
2024	8,784	0.332	0.315	0.135	0.121	0.411
2025	8,760	0.281	0.265	0.167	0.165	0.362
2026	7,296	0.254	0.246	0.169	0.172	0.347
ETH/USDT						
full sample	42,315	0.244	0.237	0.123	0.126	0.285
zero-fee window	6,168	0.265	0.254	0.098	0.095	0.253
standard fees	36,147	0.244	0.237	0.127	0.131	0.291
2022	8,718	0.283	0.275	0.094	0.091	0.211
2023	8,757	0.246	0.228	0.118	0.119	0.331
2024	8,784	0.319	0.307	0.129	0.118	0.336
2025	8,760	0.250	0.241	0.132	0.136	0.296
2026	7,296	0.212	0.199	0.147	0.163	0.270

Ceiling: in-sample R^2 of observed hourly taker imbalance on 50 quantile-bin means of the standardised bar return, the upper bound for any return-based classifier calibrated to that period. corr^2(BVC): squared correlation of hourly BVC with observed imbalance. Slope: OLS slope of observed imbalance on BVC (its calibration). SD imb.: standard deviation of observed hourly imbalance. Ceiling (1m): the same bound for one-minute imbalance on one-minute returns.

Table 15: BVC accuracy by regime, BTC/USDT

		Pearson		Same sign		
Split	Group	1h	24h	1h	24h	Hours
volatility	high vol	0.499	0.619	0.715	0.744	13,119
volatility	low vol	0.493	0.432	0.717	0.649	15,252
volatility	mid vol	0.503	0.507	0.715	0.690	13,368
year	2022	0.484	0.580	0.721	0.720	8,718
year	2023	0.503	0.497	0.728	0.661	8,758
year	2024	0.561	0.595	0.728	0.701	8,784
year	2025	0.515	0.544	0.707	0.676	8,760
year	2026	0.496	0.533	0.690	0.709	7,296
fee regime	standard fees	0.507	0.524	0.710	0.693	36,148
fee regime	zero-fee promo	0.600	0.595	0.749	0.690	6,168

Volatility terciles use the 24h Garman-Klass volatility ranked against the trailing 30 days.

Table 16: Rolling six-month estimation window, BTC/USDT

Model	Expanding R^2_{OOS}	Rolling R^2_{OOS}	Rolling CW p	Rolling IC
EN: TI	−0.055	−0.411	0.52	0.001
EN: MS_{proxy}	−0.139	−0.379	0.73	−0.007
EN: MS	−0.006	−0.400	0.55	−0.002
EN: ALL	−0.025	−0.433	0.33	−0.006

Out-of-sample R^2 (%) against the zero forecast with the expanding window of the main analysis and with a six-month rolling window; Clark-West p-value and IC for the rolling window.

Table 17: Drop-one-group ablation, full elastic net, BTC/USDT

Group dropped	Features	R^2_{OOS} (%)	IC
none (EN: ALL)	45	−0.025	0.019
spread impact	41	−0.115	−0.004
order flow	40	−0.072	0.014
volatility	39	0.021	0.018
efficiency structure	39	−0.169	0.006
activity	43	−0.045	0.011
trade level	37	−0.125	0.005
intrahour 1m	41	−0.118	0.017

Table 18: Conditional out-of-sample accuracy, BTC/USDT

Split	Group	Hours	EN: TI	EN: MS	EN: ALL	EN: $\text{Flow}_{\text{taker}}$
vol regime	high	11,647	−0.150 (0.38)	−0.029 (0.47)	−0.080 (0.36)	−0.033 (0.72)
vol regime	low	13,914	0.042 (0.05)	−0.056 (0.46)	0.054 (0.06)	−0.087 (0.73)
vol regime	mid	12,032	0.035 (0.05)	0.085 (0.03)	0.006 (0.12)	0.005 (0.25)
illiq regime	high	12,524	−0.049 (0.17)	−0.026 (0.39)	0.056 (0.12)	0.002 (0.32)
illiq regime	low	13,471	0.103 (0.03)	0.070 (0.02)	0.024 (0.08)	−0.043 (0.67)
illiq regime	mid	11,598	−0.254 (0.64)	−0.074 (0.65)	−0.181 (0.55)	−0.072 (0.85)
session	Asia 00-08	12,528	−0.166 (0.33)	−0.086 (0.58)	−0.355 (0.73)	0.002 (0.19)
session	Europe 08-16	12,533	0.014 (0.14)	0.083 (0.03)	0.134 (0.02)	−0.053 (0.83)
session	US 16-24	12,532	−0.077 (0.23)	−0.076 (0.66)	−0.026 (0.23)	−0.038 (0.61)
weekend	weekday	26,867	−0.050 (0.13)	−0.005 (0.24)	0.020 (0.05)	−0.043 (0.80)
weekend	weekend	10,726	−0.085 (0.16)	−0.010 (0.27)	−0.287 (0.52)	0.007 (0.21)
fee regime	standard fees	31,483	−0.032 (0.05)	−0.001 (0.17)	−0.026 (0.08)	−0.026 (0.52)
fee regime	zero-fee (BTC/USDT promo)	6,110	−0.137 (0.64)	−0.023 (0.92)	−0.023 (0.92)	−0.072 (0.84)

Out-of-sample R^2 against zero (%) within each group, Clark-West p-value in parentheses. Volatility and illiquidity terciles rank the 24h Garman-Klass volatility and the Amihud ratio against the trailing 30 days, so group membership is known at forecast time.

Table 19: Sub-period accuracy, BTC/USDT

Sub-period	BTC ret. (%)	EN: TI	EN: MS_{proxy}	EN: MS	EN: ALL	EN: Flow_{BVC}	EN: $\text{Flow}_{\text{taker}}$
2022-07-10–2023-08-14	36	−0.162	−0.354	−0.018	−0.019	−0.228	−0.057
2023-08-14–2024-09-08	84	−0.086	−0.035	0.045	0.034	−0.058	−0.068
2024-09-08–2025-10-05	132	0.319	−0.011	−0.050	−0.079	−0.008	−0.016
2025-10-05–2026-10-31	−38	−0.272	−0.137	−0.009	−0.052	0.000	0.012

Out-of-sample R^2 against zero (%) in four equal sub-periods of the test sample, with the BTC return over each sub-period.

Table 20: Out-of-sample forecast accuracy, ETH/USDT

Model	R^2_{OOS} vs. 0 (%)	R^2_{OOS} vs. mean (%)	CW t vs. 0	IC	95% CI	Hit rate	Hit t
Historical mean	−0.083	0.000	–	−0.009	[−0.028, 0.008]	–	–
EN: TI	0.102	0.186	2.97***	0.012	[−0.003, 0.029]	0.499	−0.20
EN: MS_{proxy}	−0.266	−0.182	−1.60	−0.005	[−0.022, 0.011]	0.495	−0.83
EN: MS	−0.335	−0.252	−1.72	0.003	[−0.015, 0.020]	0.497	−0.46
EN: ALL	−0.356	−0.273	−1.63	−0.007	[−0.025, 0.010]	0.491	−1.41
EN: Flow_{BVC}	−0.083	0.001	−0.34	−0.014	[−0.031, 0.003]	0.491	−1.91
EN: $\text{Flow}_{\text{taker}}$	−0.010	0.073	1.06	0.009	[−0.008, 0.026]	0.505	1.08
GBM: MS	−0.441	−0.357	−0.32	0.002	[−0.016, 0.017]	0.501	0.25
GBM: ALL	−0.478	−0.394	−0.62	−0.016	[−0.034, 0.001]	0.489	−2.93

Table 21: Incremental predictive information, ETH/USDT

Model	Benchmark	Nested	R^2_{OOS} rel. (%)	CW t	CW p	DM t	DM p
EN: ALL	EN: TI	yes	−0.459	−1.09	0.861	−2.90***	0.004
EN: MS	EN: TI	yes	−0.438	−0.87	0.808	−2.64***	0.008
EN: MS	EN: MS_{proxy}	yes	−0.069	0.37	0.354	−0.71	0.478
EN: MS_{proxy}	EN: TI	yes	−0.369	−0.47	0.682	−2.87***	0.004
GBM: ALL	EN: TI	yes	−0.580	−0.36	0.639	−2.70***	0.007
EN: $\text{Flow}_{\text{taker}}$	EN: Flow_{BVC}	no	0.072	–	–	1.30	0.194

Table 22: Return response to order-flow shocks, ETH/USDT

Regressor (1 SD)	$k = 1$h	$k = 2$h	$k = 4$h	$k = 8$h	$k = 12$h	$k = 24$h
Observed taker imbalance	0.10	−0.53	0.55	0.61	0.90	2.68
	(0.33)	(−1.14)	(0.76)	(0.53)	(0.58)	(1.21)
BVC imbalance	0.56	0.81	1.85	3.62	4.58	3.67
	(1.10)	(1.10)	(1.74)	(2.29)	(2.36)	(1.31)
Observed imbalance, given z_t	−0.15	−0.79	0.21	−0.41	−0.46	2.10
	(−0.45)	(−1.55)	(0.26)	(−0.31)	(−0.26)	(0.81)
Bar return z_t, given imbalance	0.58	0.62	0.80	2.39	3.21	1.36
	(1.10)	(0.85)	(0.76)	(1.61)	(1.69)	(0.51)

Table 23: Proxy validation, ETH/USDT

	Pearson	Spearman	Same sign	N
Panel A. Signed order flow: proxy vs. observed taker imbalance				
BVC, 1h bars, per hour	0.487	0.532	0.708	42,316
BVC, 1h bars, 4h sums	0.470	0.530	0.708	42,310
BVC, 1h bars, 24h sums	0.513	0.561	0.720	42,270
BVC, 1m bars summed to 1h	0.556	0.605	0.735	42,359
Tick rule, 1h bars, per hour	0.432	0.418	0.708	42,357
Panel B. Price impact and volatility				
Kyle λ (48h): BVC vs. observed flow	0.440	0.503	–	41,978
log GK vol (1h bars) vs. log RV (1m), 24h	0.987	0.984	–	41,978
Panel C. Information ceiling of return-based classifiers (out-of-sample R^2)				
	BVC raw	BVC rescaled	Any $f(z)$	N
1h bars	−8.868	0.218	0.222	21,158
1m bars	−0.153	0.276	0.286	1,270,544
Panel D. Spread estimators (one tick = 0.043 bps)				
	Median (bps)	Mean (bps)	Share = 0	Corr. log RV
Corwin-Schultz (24h mean)	20.7	23.2	0.0%	0.85
Abdi-Ranaldo (24h)	0.0	11.1	54.7%	0.61
Roll (48h)	17.6	24.0	40.6%	0.52

Table 24: BVC accuracy by regime, ETH/USDT

		Pearson		Same sign		
Split	Group	1h	24h	1h	24h	Hours
volatility	high vol	0.500	0.585	0.714	0.745	13,547
volatility	low vol	0.478	0.468	0.697	0.699	15,050
volatility	mid vol	0.510	0.541	0.715	0.717	13,142
year	2022	0.525	0.666	0.713	0.758	8,718
year	2023	0.478	0.462	0.700	0.678	8,758
year	2024	0.554	0.575	0.727	0.733	8,784
year	2025	0.491	0.541	0.703	0.725	8,760
year	2026	0.446	0.477	0.693	0.701	7,296
fee regime	standard fees	0.487	0.514	0.708	0.723	36,148
fee regime	zero-fee promo	0.504	0.571	0.710	0.700	6,168